

CST Third Semester Final Examination, December 2023
Signals and Systems (CS-2104)

Plan-A: Answer any 5 out of 7 questions, $3 + 3 + 4 = 10$ marks.

Plan-B: Answer any 15 subparts out of the 21 subparts of 7 Qs.

Full Marks 50

Time 180 minutes

1. (a) How is frequency modulation different from amplitude modulation?
(b) Describe design of frequency demodulators and explain how this works.
(c) How does the frequency modulation happen when the carrier is a square wave with frequency ω_c and signal frequency is bandlimited to ω_m ? Sketch the time domain waveform and the frequency spectrum of such pulse frequency modulation for proper explanation.
2. A third order dynamical system is described by the differential equation :
$$\ddot{x} + 9\dot{x} + 23x + 15x = y \text{ i.e. } \frac{d^3x}{dt^3} + 9\frac{d^2x}{dt^2} + 23\frac{dx}{dt} + 15x = y$$
 - (a) Obtain state space representation of the system with state variables one derivative apart.
 - (b) Obtain the poles and zeroes for this system.
 - (c) Obtain the state transition matrix for this system by solving the equation.
3. (a) Explain mathematically the effect of time delay in Laplace transform domain.
(b) Find and sketch roughly the impulse response of the following system:
 $2y[n] - \frac{1}{a}y[n-1] = x[n]$. Value to be used is $a = 2$ for even roll numbers and $a = 3$ for odd roll numbers.
(c) Explain the concept used for plotting the magnitude and phase in frequency domain for the output $Y(j\omega)$ described in terms of input $X(j\omega)$ and system $H(j\omega)$ in the transform domain.
4. Write short notes on:
 - (a) Opamp based realization of proportional, integral and derivative actions
 - (b) Amplitude modulation and demodulation for communication
 - (c) Frequency selective filter design
5. Describe briefly the terms and explain their trade-offs, as applicable:
 - (a) Time division multiplexing and Frequency division multiplexing in communication systems
 - (b) Narrowband and Wideband frequency modulation
 - (c) Time domain and frequency domain response of higher order filters
6. (a) Explain how Z transform emerges from ideal sampling of continuous signal.
(b) State and prove the final value theorem in Z domain and hence comment on how it could be useful to obtain the inverse Z transform.
(c) Explain why the stable region of Z-plane falls within the unit circle while the stable region of s-plane lies in the left half of the $j\omega$ axis by mapping from one complex plane to other.
7. (a) The open loop transfer function of a system is $H(s) = \frac{b}{(s-a)}$ and a proportional feedback with gain $G(s) = K$ is used to control the system. Suggest values of K so that the system becomes stable.
(b) Explain the phenomenon of destabilization due to positive feedback.
(c) How is feedback control achieved through observation and measurement? Describe with explanatory block diagrams.